

Code Layout, Readability & Reusability

Project: Rising Water – AI-Based Flood Prediction System

1. Project Structure

The project follows a modular folder structure:

- app.py
- dataset/
- models/
- templates/
- static/
- Project Documentation/

2. Coding Standards

- Meaningful variable and function names
- Consistent indentation (PEP 8)
- Modular functions
- Proper exception handling
- Readable code formatting

3. Readability

The application is divided into frontend, backend, model, and dataset components. Clear naming conventions and comments improve readability and maintenance.

4. Reusability

Component	Reuse
Flask Routes	Can be extended with additional pages
XGBoost Model	Can be retrained with new datasets
Scaler	Reusable for future predictions
HTML Template	Easy to customize
CSS	Reusable across pages

5. Best Practices

- Separate frontend and backend
- Store trained model in models folder
- Keep reusable styling in CSS
- Document code clearly
- Use version control with GitHub

6. Conclusion

The Rising Water project is organized using a clean and maintainable architecture, making the code easy to understand, reuse, and extend.